

Sai Nikshitha Mastha

Hyderabad, Telangana, India

+91-8885433496 — sainikshithamastha396@gmail.com — GitHub — LinkedIn

Objective

Final-year Computer Science undergraduate with hands-on experience developing AI-enabled engineering solutions using LLM APIs (OpenAI, Claude, Gemini), prompt engineering, structured outputs, and multi-agent systems. Experienced in ML model development and serving with Hugging Face, XGBoost, and CNNs, alongside backend engineering using Python, Node.js/Express, REST APIs, PostgreSQL/MySQL, and MongoDB. Strong focus on automating workflows, integrating AI into software systems, and building reliable end-to-end solutions from prototype to deployment.

Technical Skills

AI & LLM Engineering: LLM API Integration (OpenAI, Claude, Gemini), Prompt Engineering, Structured/JSON Outputs, LangChain, Hugging Face Transformers, Model Endpoints, AutoGen (Multi-Agent Systems), Retrieval-Augmented Generation (RAG), AI Workflow Automation, Model Evaluation

Machine Learning: XGBoost, CNN, Scikit-learn, Pandas, NumPy, Feature Engineering, EDA, Image Preprocessing & Augmentation, Model Validation

Backend & APIs: Python, Node.js, Express.js, REST API Design, Third-Party API Integration, JWT Authentication, Request Validation, Error Handling, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, SQL, Joins, Indexing, Schema Design, Transactions

Cloud & Developer Tools: AWS, Docker, Git, GitHub, Jenkins, Maven, Postman.

Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS.

Education

Keshav Memorial Institute of Technology
Bachelor of Technology in Computer Science

2023 – 2027
CGPA: 9.0

Sri Chaitanya Junior College
Intermediate (MPC)

2021 – 2023
Percentage: 97.5

Nalanda High School, Nizamabad
Secondary School Certificate (SSC)

2020 – 2021
GPA: 10

Projects

AgentMonitor: Predictive and Secure Multi-Agent Systems

Python — XGBoost — AutoGen — Live Demo

- Self-initiated and built a predictive monitoring framework for **multi-agent AI systems** in Python, applying **machine learning classification (XGBoost)** to detect anomalous agent behavior in near real time.
- Engineered feature pipelines from agent telemetry data and built **Human-in-the-Loop checkpoints** into the architecture, treating agentic-system security and auditability as first-class design concerns.
- Deployed and running live at agentmonitor-lvi7.onrender.com, beyond a toy/local-only project.

Enterprise Data Privacy Anonymization Pipeline

Node/TypeScript — LLM APIs — PostgreSQL

- Built a data-classification and anonymization pipeline integrating **third-party LLM APIs** (Claude, Gemini, OpenAI) to classify data sensitivity and generate anonymization rules (masking, pseudonymization, tokenization), consuming **structured JSON** responses directly in pipeline logic.
- Designed and queried a **PostgreSQL** database to store structured datasets, sensitivity classifications, and audit trails, with schema decisions made to support compliance/audit reporting.
- Collaborated with a teammate and a Salesforce engineer mentor via a shared Git repository and pull-request workflow, as part of the **Salesforce Compass Program** on an enterprise-grade platform project.

AI-Driven Plant Disease Detection System

Express.js — Node.js — REST API — Third-Party Model Integration

- Built an **Express/Node.js REST API** backend with request validation and error handling, integrating a **CNN model served via a third-party endpoint (Hugging Face)** for real-time image classification.
- Preprocessed and augmented image datasets** and evaluated model accuracy against validation splits to validate correctness of AI-generated predictions before they reached the API layer.
- Designed a MySQL relational schema alongside MongoDB for session/history data, evaluating normalized vs. denormalized trade-offs for real read/write patterns.

College Canteen App

Node.js — Express — JWT Auth — MongoDB — WebSockets — Docker

- Architected a multi-tier backend with **JWT-authenticated REST APIs** and **WebSockets** for real-time order-status sync between two distinct user roles (students and canteen admins), a system with real, active users.
- Used **SQL transactions** to keep concurrent order-status updates atomic and consistent, reasoning explicitly about data-integrity edge cases.
- Containerized** the backend with **Docker** on a **Unix-based** environment for consistent, repeatable deployment.

Achievements

- Solved **200+ Data Structures & Algorithms problems** on LeetCode, with strong command of complexity analysis and problem decomposition.
- Selected for the **Salesforce Compass Program** – one of a limited cohort mentored directly by a Salesforce software engineer on an enterprise-grade platform project, collaborating in a distributed setup.
- Active member of **Google Developer Groups (GDG) Campus** – technical workshops and hackathons, contributing to and building alongside a developer community.